

Techno-economic and multicriteria analysis of grid-connected energy systems for hydrogen production: A case study from Bangladesh

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ABSTRACT

This study evaluates the feasibility of grid-connected hybrid renewable energy systems (HRES) for green hydrogen production in Cox's Bazar, Bangladesh—a region with favorable climate conditions for renewable energy generation. Hydrogen, as a clean energy carrier, offers significant potential to reduce fossil fuel dependency and support sustainable transportation infrastructure. Unlike existing studies focusing on general feasibility in various countries, this research uniquely addresses hydrogen generation from renewables and its techno-economic aspects in Bangladesh, where hydrogen systems remain unexplored despite numerous HRES electrification projects. A multicriteria decision-making (MDCM) approach was adopted, combining the Criteria Importance Through Intercriteria Correlation (CRITIC) method to determine criterion weights and Technique for Order Preference by Similarity to Ideal Solution (TOPSIS) to rank four proposed HRES configurations from 10 sub-criteria. Social criteria, including job potential and the Human Development Index (HDI), were integrated alongside economic, technical, and environmental criteria to provide a comprehensive evaluation. The CRITIC ranking identifies HRES3, integrating PV, wind turbines, batteries, and grid connectivity, as the optimal solution with an LCOE of \$0.0321/kWh, NPC of \$10.11 M, 91.7% renewable fraction, LCOH of \$5.22/kg, HDI of 0.62, and job potential of 24.90 over 25 years. Emission analysis reveals that HRES3 produces significantly lower CO₂ emissions compared to other configurations. Sensitivity analysis shows that HRES3 maintains stable LCOE and NPC, despite variations in hydrogen load, nominal discount rate, and CO₂ penalties, which emphasizes its resilience and economic stability under changing conditions.

1. Introduction

The worldwide shift toward sustainable energy systems is gaining momentum as both governments and industries increasingly prioritizing climate change solutions. The urgency to reduce greenhouse gas emissions has sparked growing interest in renewable energy and clean fuel options, with hydrogen emerging as a key player in the future of energy. Hydrogen is viewed as a clean, adaptable fuel that can help lower carbon emissions in sectors difficult to decarbonize, such as heavy industry, transportation, and power, especially when derived from HRESs [1]. It stands out for its low environmental impact and high energy efficiency, with a weight-based energy density surpassing many conventional fuels. This makes it an attractive clean energy source, particularly for fuel cell

vehicles, which have seen significant development. Research has shown that green hydrogen fuels significantly impact the transportation sector and help reduce greenhouse gas emissions [2]. Hydrogen can be produced from various energy sources that offers flexibility in production methods. Using renewable energy further enhances its environmental benefits, positioning hydrogen as a crucial component in the shift to sustainable energy systems. Advancing technology positions hydrogen as key to decarbonizing transport, industry, and power for sustainability [3]. Globally, hydrogen is gaining recognition as essential for the clean energy shift. Its environmental impact depends on the production method. Currently, steam methane reforming technique, which produces a significant amount of CO₂, is used to manufacture most of the hydrogen from natural gas [4]. In contrast, green hydrogen is produced through water electrolysis powered by HRESs, offers a carbon-free

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Abbreviations

Y_{rated}	Rated capacity of the PV array (kW)	C_h	The total capital and operational expenses related to hydrogen production
D_f	PV derating factor (%)	C_e	The cost of electricity used during electrolysis
G'	Solar radiation incident on the PV array in the current time step (kW/m ²)	LCOE	The levelized cost of electricity
G'_{stc}	Incident radiation at standard test conditions (kW/m ²)	E_i	The total electricity consumption in the i -th year (kWh)
α_p	Temperature co-efficient of power (%/°C)	C_{el}	The unit cost of the electrolyzer
T_{cell}	PV cell temperature in the current time step (°C)	P_e	The annual power consumption required for electrolysis (kW)
T_{stc}	PV cell temperature under standard test conditions (°C)	CF	The capacity factor, indicating the system's utilization rate
$\tau\alpha$	Effective transmittance and absorptance of the PV array (%)	η	The efficiency of the electrolyzer
R_T	Solar radiation incident on the PV array (kW/m ²)	E_Y	Yearly electricity served (kWh)
η_{con}	Electrical conversion efficiency of the PV array	N	Number of years
H	Coefficient of heat transfer to the surroundings (kW/m ² °C)	p	Real annual interest rate (%)
T_{amb}	Ambient temperature, both measured (°C)	p'	Nominal interest rate (%)
$T_{\text{cell,NOCT}}$	Nominal operating cell temperature (°C)	S	Yearly inflation rate (%)
$T_{\text{amb,NOCT}}$	Ambient temperature (20°C)	E_T	Total capacity shortage (kWh/yr)
$R_{T,NOCT}$	Solar radiation at 20 °C, (0.8kW/m ²)	E_U	Total unmet load (kWh/yr)
V_{hub}	Wind speed at the hub height of the wind turbine (m/s)	E_D	Total annual electrical demand (kWh/yr)
V_{anem}	Wind speed at anemometer height (m/s)	E_P	Total electricity production (kWh/yr)
H_{hub}	Hub height of the wind turbine (m)	HDI	Human development index
H_{anem}	Anemometer height (m)	E_{Load}	Total annual AC load of the community (kWh/yr)
H	Surface roughness length (m)	E_{ex}	Annual excess energy produced by the system (kWh/yr)
$P_{\text{out,wtg}}$	Wind turbine power output (kW)	F_{ex}	Factor for maximum excess energy useable by additional loads
P_{stp}	Wind turbine power output at Standard Temperature and Pressure (kW)	F_{ld}	Factor for maximum load allowable for new additional loads
ρ_{actual}	Actual air density (kg/m ³)	job_x	Employment factor (jobs per unit of energy generated) for a specific component x
ρ_{stp}	Air density at standard temperature and pressure (1.225kg/m ³)	$E_{\text{gen},x}$	Energy generation contribution (e.g., kWh/year) of component x
Q_a	Available energy at the beginning of the time step (kWh)	HHV _{H₂}	Heating value of hydrogen fuel (MJ/kg)
Q_b	Total stored energy at the beginning of the time step (kWh)	P_{EZ}	DC power of the electrolyzer
s	Storage capacity ratio	m_{H_2}	Mass flow rate hydrogen generated to the electrolyzer (kg/s)
z	Storage rate constant (h ⁻¹)	η_{H_2}	Efficiency of the electrolyzer
T_L	Duration of the time step (h)	R_{hydrogen}	Hydrogen production rate
β	Storage's maximum charging rate (A/Ah)	I_{ele}	Current of the electrolyzer
Q_m	Overall storage bank capacity (kWh)	η_F	Faraday efficiency
N	Number of batteries in the storage bank	N_c	Total number of cells in series
I	Storage's maximum charging current (A)	F	Faraday's coefficient
V	Storage's nominal voltage in (V)	N_p	Number of people in the community
η_s	Storage charge efficiency	P_{FC}	Power generated by fuel cell
Q_m	Overall storage bank capacity (kWh)	N_S	Number of cells in the stack
η_d	Storage discharge efficiency	V_{FC}	Voltage output of the fuel cell
$P_{\text{out,con}}$	Output power of the inverter (kW)	I_{FC}	Current of fuel cell
η	Efficiency of inverter	E	Fuel cell internal voltage
$P_{\text{in,con}}$	Input power of the inverter (kW)	V_{act}	Activation voltage
C_{AEC}	Total annual energy charge	V_{con}	Concentration voltage
$E_{\text{gp},x,y}$	Energy procured from the grid (kWh)	V_{Ω}	Ohmic voltage
$E_{\text{gs},x,y}$	Energy sold to the grid (kWh)	F_1	Set of benefit criteria (higher values are better)
$P_{\text{power},x}$	Price of grid power corresponding to rate x (/kWh)	F_2	Set of cost criteria (lower values are better)
$P_{\text{sellback},x}$	Rate at which electricity is sold back for rate x (/kWh)	a_{ij}	Normalized value for the i^{th} alternative and j^{th} criterion
$P_{\text{pgd},x,y}$	Peak hourly grid demand in month y while rate x is in effect (kWh)	w_j	Weight for the j^{th} criterion
D_x	The rate of grid demand for rate x (/kW/month)	r_{ij}	Normalized value for alternative i and criterion j
C_C	Capital cost on a yearly basis	x_{ij}	Original value for alternative i and criterion j
C_R	Cost of replacement over a year	x_j^{max}	Maximum value of criterion j
C_{OM}	Annualized cost of operation and maintenance	x_j^{min}	Minimum value of criterion j
C_f	Annualized cost of fuel	$\bar{r}_j$	Mean of normalized scores for criterion j
S	Annualized salvage value	m	Total number of alternatives
T	The operational lifespan of the electrolyzer, expressed in years	r_{jk}	Correlation coefficient between criteria j and k
H_i	The total amount of hydrogen produced annually,	n	Total number of criteria
	measured in kilograms	σ_j	Standard deviation of criterion j

$\sum_{k=1}^n (1 - r_{jk})$	Measure of conflict for criterion j
C_j	Information content of criterion j
w_j	Objective weight of criterion j
C_i	Closeness coefficient for the i^{th} alternative

S_i^+	Distance of the i^{th} alternative from the ideal solution
S_i^-	Distance of the i^{th} alternative from the negative ideal solution

alternative that aligns with global net-zero emissions targets set for 2050. The European Union, Japan, and South Korea have adopted robust hydrogen strategies as part of their climate goals, emphasizing the need for similar initiatives in Bangladesh [5].

Bangladesh is poised for a hydrogen energy shift, with renewable investments and feedstock resources. Two key initiatives have been launched to investigate hydrogen production from water and biomass. The Bangladesh Council for Scientific and Industrial Research (BCSIR) in Chittagong is leading efforts in hydrogen production through photocatalytic water splitting and biomass gasification. While biomass gasification is still in the pilot phase, photocatalytic hydrogen production is currently at the experimental stage. Furthermore, since 2021, another project has been dedicated to producing hydrogen from biogas using reforming technology, which is also undergoing laboratory-scale evaluation [6]. Technological barriers to hydrogen energy adoption have significantly diminished due to the global push for cleaner energy and the depletion of fossil fuel reserves. However, challenges remain, especially concerning grid power reliability, as unpredictable weather can affect renewable energy generation. Key economic challenges include high capital expenditures for HRESs, operational costs for system maintenance and electrolyzers. In Bangladesh, the cost of solar PV systems has substantially declined, making solar energy one of the country's most cost-effective electricity sources. Although wind energy remains more expensive, it is expected to become more affordable with technological advancements. Biomass, primarily sourced from agricultural residues, can further support HRESs, helping stabilize energy supply and enhance sustainability [7,8].

Numerous studies globally have assessed the use of HRESs for hydrogen production, highlighting the economic and environmental advantages of renewable hydrogen. These studies show that combining solar, wind and storage technologies can produce hydrogen at a competitive cost compared to traditional fossil fuel-based methods, especially in regions with strong renewable resources.

For instance, a study in Ref. [9] developed a PV/Wind/Fuel Cell system with a power output of 3.8 kW, where excess energy was used to produce hydrogen for storage. Similarly, the viability of a stand-alone PV/Diesel/Battery/Fuel Cell system was analyzed in Ref. [10] using HOMER, revealing the potential of hydrogen to replace diesel generators and reduce emissions. Research in Ref. [11] evaluated a PV/Fuel Cell/Battery system for a remote Brazilian community, focusing on initial costs, NPC and LCOE with hydrogen storage integration. Several studies have also explored the cost-effectiveness of integrating multiple storage and energy generation technologies. Research conducted in Ref. [12] examined a PV/Wind/Fuel cell system for off-grid vacation homes in Turkey, analyzing two occupancy scenarios (regular and seasonal) and two storage options (battery and hydrogen), with findings indicating that Battery storage was more cost-effective than Hydrogen. An economic analysis of a Wind/PV/Diesel system in Iran was conducted in Ref. [13] identifying the Wind/Hydrogen/Battery system as the most profitable. In another study [14], the technical and economic prospects of a hydrogen-based electric vehicle charging station in Delhi were assessed, concluding that a hydrogen-powered HRES would be a profitable solution for the region, promoting sustainable transportation. Other studies have explored geographical and market-specific challenges. For example [15], focused on electricity and hydrogen production across three islands in Eastern Canada, finding that the LCOE was highly sensitive to fluctuations in fuel market costs. In Ref. [16], a techno-economic analysis of a methanol plant revealed that reducing PV

and wind turbine costs by 50% could make hydrogen-based electricity production more affordable than traditional methods. The use of MCDM approaches in HRES studies has also been highlighted. In Ref. [17], HOMER was used alongside MCDM methods to evaluate energy, economic, and environmental performance in northern Ghana, including the potential for hydrogen production and job creation. Similarly, research in Ref. [18] compared Wind and Wind/PV scenarios on Bozcaada Island, Turkey, demonstrating significant cost reductions with HRES. Studies such as [19,20] assessed the feasibility of PV/Wind energy systems for hydrogen refueling stations in China and Turkey, respectively, showing that Wind/PV/Battery configurations had lower LCOE values. Globally, HRESs have been explored to address specific regional challenges. In South Africa [21], identified coastal sites as more favorable for wind-powered hydrogen refueling stations, with production costs ranging between \$6.34 and \$8.97 per kilogram. In Canada [22], assessed an off-grid PV system with hydrogen storage using underwater hydrogen balloons, designed to meet the energy needs of 1500 homes. A study in Ref. [23] analyzed a standalone solar PV and hydrogen fuel cell system for domestic power with heating applications, achieving an LCOE of \$0.8399/kWh. Large-scale systems have also been examined for their potential to combine different renewable technologies. In Ref. [24], a standalone system incorporating wind, bifacial, and floating PV was evaluated for hydrogen and electricity production in remote communities. Research in Ref. [25] highlighted the applicability of PV/Wind systems for hydrogen production on Turkish islands, while [26] analyzed the feasibility of green hydrogen production in Oman, identifying solar energy as the most cost-effective option, with an LCOH ranging from \$8.34 to \$16.73 per kilogram. In Pakistan [27], demonstrated the viability of an HRES combining PV panels, fuel cells, and biogas generators to provide stable power to remote villages, achieving an LCOE of \$0.41/kWh and an NPC of \$199,081. Similarly, in Dakhla City [28], evaluated a hybrid microgrid incorporating wind, diesel, fuel cells, and electrolyzers, finding it economically viable with a leveled energy cost of \$0.0701/kWh and the potential to reduce CO₂ emissions by 70,529.80 tons over 20 years. In Muscat [2], assessed three hybrid systems for hydrogen production and refueling stations, identifying the PV/Wind/Battery system as the most cost-effective option with the lowest NPC, LCOE, and LCOH. In the Middle East [29], conducted a quantitative analysis of green hydrogen generation, focusing on electricity production, costs, hydrogen yield, and water requirements. The study found that Egypt and Yemen have the best potential for green hydrogen production using solar energy, with wind resources in Egypt and Oman also offering competitive LCOH. Projections indicated that solar energy could drive LCOH below \$1/kg by 2050.

A review of the literature reveals that, while many studies have explored feasibility in various countries, none have specifically addressed hydrogen generation from renewables or techno-economic aspects in Bangladesh, where, despite several electrification projects based on HRES, hydrogen systems have yet to be implemented. Therefore, this paper aims to evaluate the feasibility of cost-effective green hydrogen production in Bangladesh, with Cox's Bazar as the study location. Key contribution include.

- The feasibility of four grid-connected HRES for green hydrogen production is evaluated through HOMER optimization, focusing on underexplored techno-economic and social dimensions.
- An MDCM framework is introduced, using the CRITIC method for weighting and TOPSIS for ranking, ensuring a balanced evaluation

across ten sub-criteria under economic, technical, environmental, and social criteria.

- Social indicators, including job creation and the HDI, are incorporated to assess the broader societal impact.
- A sensitivity analysis is conducted to evaluate the economic impacts of hydrogen load variability, CO₂ penalties, and nominal discount rates.

The rest of the paper is structured as follows: Section 2 outlines the methodology for designing HRESs and Section 3 presents results analysis

from simulation. Finally, section 4 concludes with key findings.

2. Methodology

This paper follows an optimized framework for a HRES that can meet the community's load demand and facilitate green hydrogen production. The HRES design incorporates techno-economic considerations and a structured hybrid configuration. As shown in Fig. 1, the framework comprises three phases: preliminary assessment, simulation, and decision-making. In the preliminary phase, Cox's Bazar, Bangladesh,

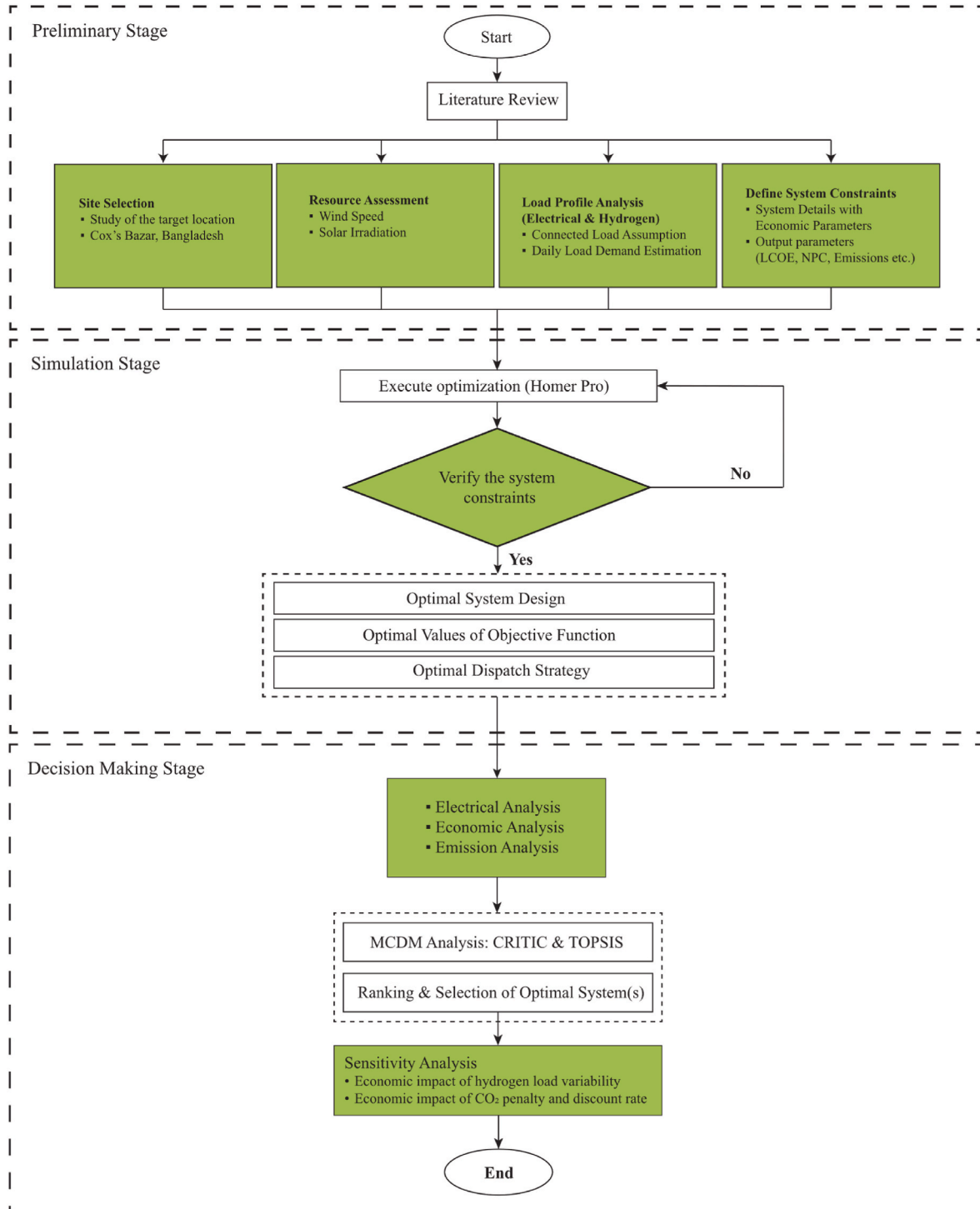


Fig. 1. Methodological approach of the system design and analysis [30].

was chosen for assessment, with meteorological data collected. A synthetic load profile for winter and summer seasons was created to estimate energy needs. The model, built using HOMER Pro, integrated cost, technical specifications, and system economics to optimize the design. Load following (LF) and cycle charging (CC), both strategies were tested to minimize LCOE and NPC. After the initial modeling, a detailed techno-economic assessment was performed using HOMER Pro, evaluating component sizing, energy generation, costs, emissions, and financial performance. In the decision-making phase, the optimized results were subjected to a MCDM analysis to rank the systems based on economic, technical, environmental, and social criteria. The CRITIC method was used to calculate criterion weights, while TOPSIS was employed to rank the systems and identify the most optimal design. Sensitivity analysis was conducted to assess the impacts of hydrogen load variability, CO₂ penalties, and discount rates on system performance. The selected system was evaluated for its feasibility and sustainability, based on its electrical performance, economic viability, emissions reduction, and social benefits, including job creation and HDI.

2.1. Site selection and resources assessment

Cox's Bazar, located in the southeastern part of Bangladesh, is globally renowned for having the longest natural sea beach. The area (Fig. 2) has been chosen as a key location for this study due to its geographical and climatic features that make it ideal for renewable energy potential assessments. The region covers an extensive area of approximately 23.4 km² and is situated at a longitude of 92°0.4'E and a latitude of 21°25.6'N. Its coastal location, approximately 150 km south of the port city of Chattogram, ensures it experiences consistent wind patterns and high solar irradiance levels throughout the year. It has a tropical climate, which offers abundant wind and solar resources. The wind speed data and monthly average global horizontal irradiance values for solar energy were obtained from NASA's POWER database [31]. HOMER Pro, the energy modeling software used in this study, calculated the clearness index and other relevant solar parameters based on the precise latitude and longitude coordinates. Furthermore, the coastal wind speeds in Cox's Bazar have been identified as highly favorable for wind energy production. In Fig. 3 (a), monthly solar radiation averages show that March, April, and May experience the highest irradiance levels, with an annual daily average of 4.75 kWh/m². December sees the lowest irradiance, around 2.64 kWh/m²/day, while June reaches the maximum, exceeding 7.89 kWh/m²/day. Meanwhile, Fig. 3 (b) illustrates monthly wind speed trends, with June, July, and August exhibiting the greatest average speeds of 5.75 m/s, contrasting with October, which shows the lowest at 4.22 m/s.

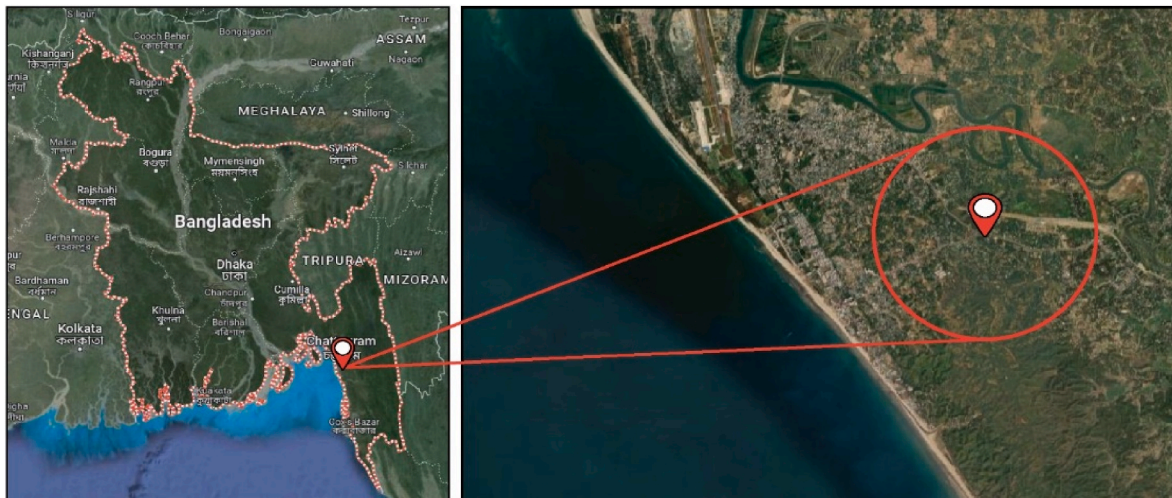


Fig. 2. Satellite image of the selected site (Cox's Bazar, Bangladesh) [32].

2.2. Load profile assessment

The hydrogen demand is met through electrolysis, powered by a HRES. HOMER Pro uses location-specific meteorological data to estimate the load profile, either daily or with random variability. The electric load, representing energy use for a community of around 6000 people, is 12,630 kWh per day, with a peak demand of 1376.5 kW and a load factor of 0.38. Peak loads, reaching up to 650 kW, occur between 10:00 a.m. and 3:00 p.m., with lower demands in the early morning and evening. The seasonal profile (Fig. 4 (a)) shows average loads between 500 kW and 1000 kW, peaking in August. A hydrogen load is incorporated to represent a local hydrogen refueling station. The hydrogen load profile (Fig. 4 (b)) shows a steady demand of 15 kg/h from 6:00 a.m. to 10:00 p.m., with minimal seasonal variation. Key metrics include an average daily load of 225 kg, a peak hourly load of 27.1 kg/h, and a load factor of 0.35. These load profiles were generated in HOMER Pro for analysis.

2.3. System modeling

Fig. 5 represents the schematic representations of four configurations of grid-connected HRES designed for hydrogen production and storage, each varying in energy source combinations. Here, HRES1 and HRES3 integrate wind energy and PV panels, along with energy storage, to supply electricity to the electric load while producing hydrogen through an electrolyzer. The hydrogen produced is retained for subsequent use in a fuel cell. In contrast, HRES2 and HRES4 rely solely on PV and energy storage, with the grid serving as a backup and excess electricity being sold back. Each proposed system is designed to generate 225 kg/day of green hydrogen, with the generated hydrogen and electricity stored to meet load demands. All systems utilize converters to manage AC and DC power flows and emphasize hydrogen production, ensuring reliability and sustainability by balancing energy loads and hydrogen supply.

2.3.1. PV system

Eq. (1) is used by HOMER to determine the PV array output [33].

$$P_{out,PV} = Y_{rated} D_f \left(\frac{G}{G_{stc}} \right) [1 + \alpha_p (T_{cell} - T_{stc})] \quad (1)$$

The PV cell's temperature indicates the array's surface temperature. While it may increase by 30 °C or more in direct sunlight, at night it is equal to the ambient temperature. HOMER calculates the cell temperature by considering both the ambient temperature and the radiation incident on the array, as shown in Eq. (2).

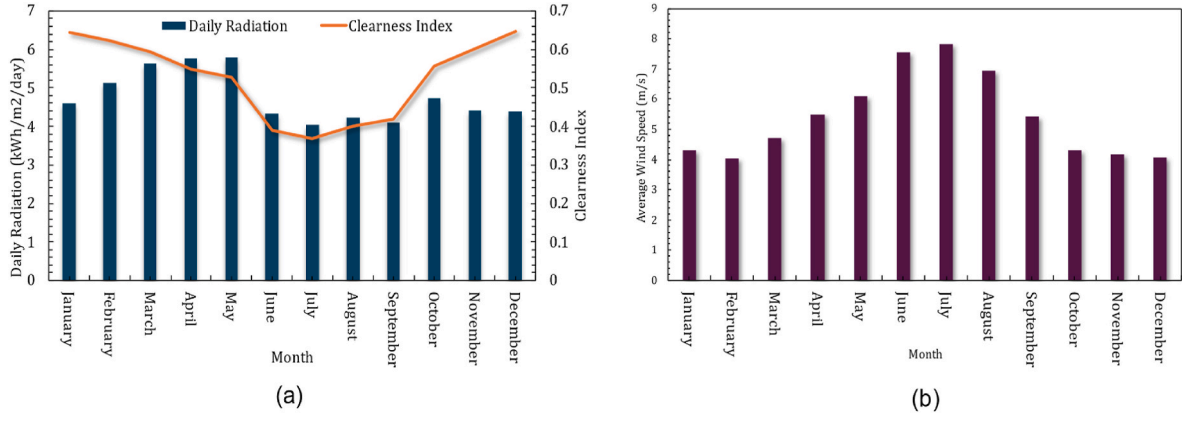


Fig. 3. Average monthly (a) solar radiation and (b) wind speed in Cox's Bazar, Bangladesh.

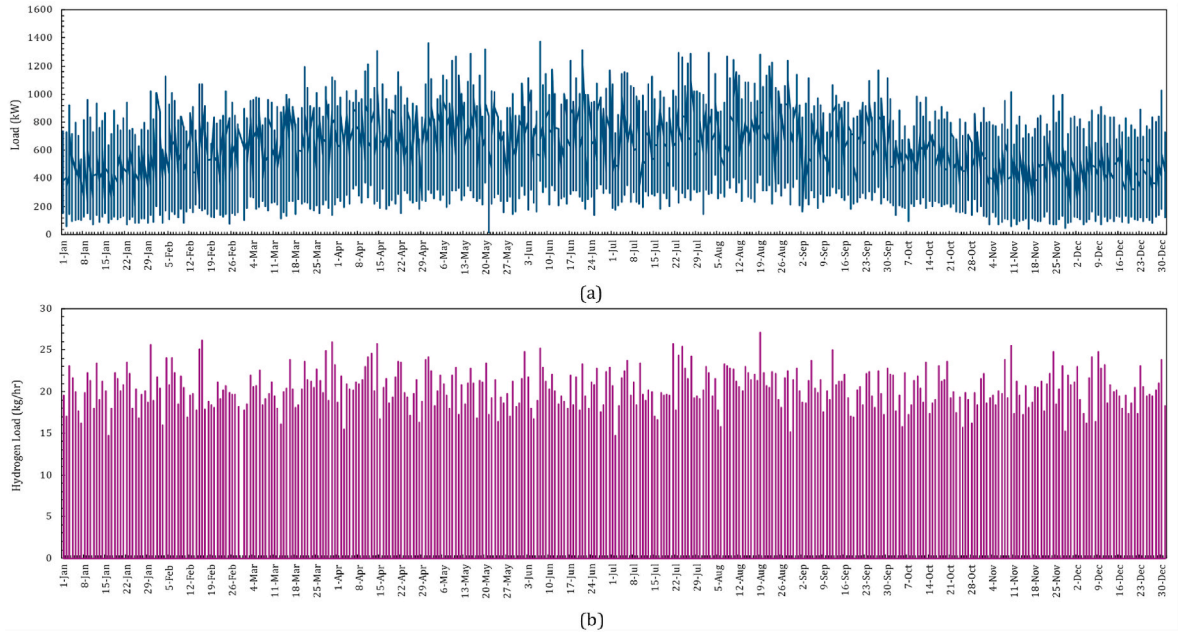


Fig. 4. (a) Electric load profile and (b) Hydrogen load profile.

$$\tau\alpha R_T = \eta_{con} R_T + H(T_{cell} - T_{amb}) \quad (2)$$

Solving Eq. (2) for the cell temperature yields Eq. (3):

$$T_{cell} = T_{amb} + R_T \left(\frac{\tau\alpha}{H} \right) \left(1 - \frac{\eta_{con}}{\tau\alpha} \right) \quad (3)$$

Because directly measuring $\left(\frac{\tau\alpha}{H}\right)$ is challenging, manufacturers typically provide the nominal operating cell temperature (NOCT) as an alternative. NOCT is determined under particular conditions: 800 W/m² of solar irradiance, an ambient temperature of 20 °C, and no-load operation (i.e., $\eta_{con} = 0$). This yields Eq. (4):

$$\left(\frac{\tau\alpha}{H} \right) = \frac{T_{cell,NOCT} - T_{amb,NOCT}}{R_{T,NOCT}} \quad (4)$$

With an estimated $\tau\alpha$ of approximately 0.9, the temperature of the PV cell can be derived Eq. (5).

$$T_{cell} = T_{amb} + R_T \left(\frac{T_{cell,NOCT} - T_{amb,NOCT}}{R_{T,NOCT}} \right) \left(1 - \frac{\eta_{con}}{0.9} \right) \quad (5)$$

2.3.2. Wind turbine

HOMER determines the wind speed at a specified hub height using Eq. (6) [33].

$$V_{hub} = V_{anem} \times \frac{\ln(H_{hub}/H)}{\ln(H_{anem}/H)} \quad (6)$$

When the power law is chosen, HOMER calculates the wind speed at the hub height using Eq. (7):

$$V_{hub} = V_{anem} \times \left(\frac{H_{hub}}{H_{anem}} \right)^\beta \quad (7)$$

Here, β represents the power law exponent. Power curves normally depict wind turbine performance under standard temperature and pressure (STP) conditions. To account for real-world conditions, HOMER adjusts the estimated power value based on the air density ratio, as shown in Eq. (8):

$$P_{out,wg} = \left(\frac{\rho_{actual}}{\rho_{stp}} \right) \times P_{stp} \quad (8)$$

2.3.3. Battery

HOMER determines how much electricity the storage bank can absorb at each time step [33]. This maximum charging capacity is then used to determine whether the storage system can accommodate all

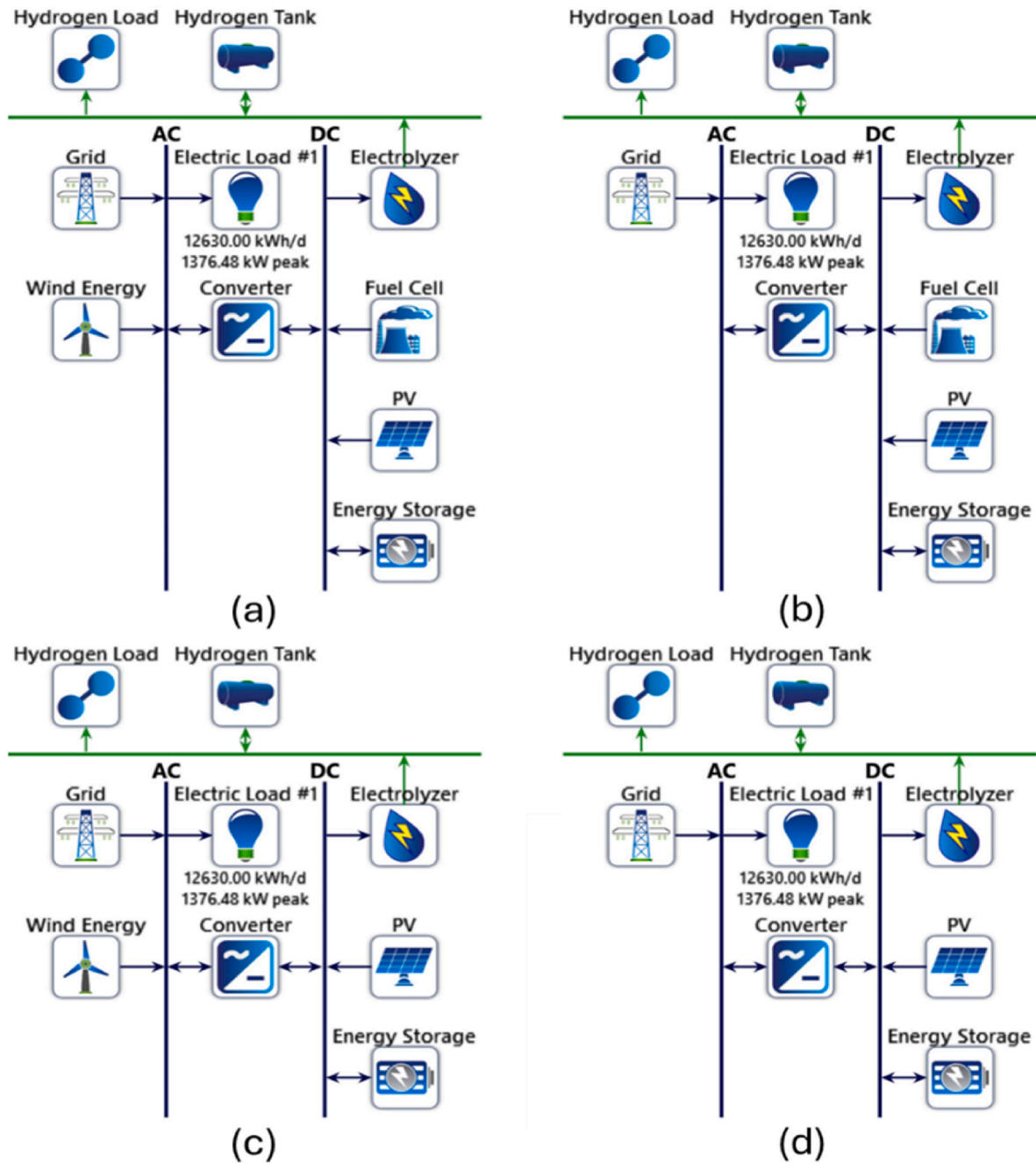


Fig. 5. Schematic diagram of the grid-connected hybrid systems (a) HRES1 (PV-Wind-Battery-Fuel cell-Grid) (b) HRES2 (PV-Battery-Fuel cell-Grid) (c) HRES3 (PV-Wind-Battery-Grid) (d) HRES4 (PV-Battery-Grid).

excess renewable energy or if an additional cyclic charging generator is required to produce extra power. Eq. (9) defines the total electricity that the two-tank storage system can hold.

$$P_{max,battery} = \frac{zQ_a e^{-zT_L} + Q_b z s(1 - e^{-zT_L})}{1 - e^{-zT_L} + s(zT_L - 1 + e^{-zT_L})} \quad (9)$$

Eq. (10) represents the power used to charge the storage system, which corresponds to its peak charging rate.

$$P'_{max,battery} = \frac{(1 - e^{-\beta T_L})(Q_m - Q_b)}{T_L} \quad (10)$$

Eq. (11) represents the maximum power available for charging the storage bank, which corresponds to the peak charging current.

$$P''_{max,battery} = \frac{NIV}{1000} \quad (11)$$

HOMER computes the maximum power for storage charging by considering the lowest value, after factoring in charging losses for each parameter. As a result, the equation is expressed as shown in Eq. (12):

$$P_{max} = \frac{MIN(P_{max,battery}, P'_{max,battery}, P''_{max,battery})}{\eta_s} \quad (12)$$

Eq. (13) defines the maximum power that the storage bank can deliver over a specific period:

$$P_{max,discharge} = \frac{-zsQ_m + zQ_a e^{-zT_L} + Q_b z s(1 - e^{-zT_L})}{1 - e^{-zT_L} + s(zT_L - 1 + e^{-zT_L})} \quad (13)$$

Using Eq. (14), the storage bank's maximum discharge power is determined [34]:

$$P_{max} = \eta_d P_{max, discharge} \quad (14)$$

2.3.4. Bidirectional converter

A bidirectional converter is used because solar PV and battery systems generate DC power but require AC power. HOMER uses Eq. (15) to calculate the converter size, considering the energy exchange between the buses [33].

$$P_{out, con} = \eta P_{in, con} \quad (15)$$

2.3.5. Electrolyzer

Water electrolysis is recognized as one of the most effective methods for hydrogen production using renewable resources. This study used a proton exchange membrane (PEM) electrolyzer for the process. The electrolyzer operates by employing an electrochemical reaction to split water into hydrogen and oxygen. The generated hydrogen is stored in a tank for subsequent use in fuel cells for electricity generation. The power demand of the electrolyzer can be described using Eq. (16) [35].

$$P_{EZ} = \frac{m_{H_2} HHV_{H_2}}{\eta_{H_2}} \quad (16)$$

The rate of hydrogen production can be calculated using Eq. (17).

$$R_{hydrogen} = \frac{I_{ele} \times \eta_F \times N_c}{2 F} \quad (17)$$

Faraday efficiency can be calculated using Eq. (18):

$$\eta_F = 96 \times \exp\left(\frac{0.09}{I_{ele}} - \frac{75.5}{I_{ele}^2}\right) \quad (18)$$

2.3.6. Hydrogen tank

Energy storage systems play a crucial role in overcoming energy shortages during nighttime and addressing seasonal variations in solar energy. Currently, surplus hydrogen is stored in a tank when production exceeds consumption. Research has shown that hydrogen storage is a more promising, cost-effective, and sustainable solution for long-term energy storage compared to conventional battery systems.

2.3.7. Fuel cell

A fuel cell is an electrochemical system that generates electricity without the need for external charging, unlike a traditional battery. It produces electrical power by converting chemical energy into electrical energy. The device features two electrodes: the anode and the cathode. The amount of power produced by the fuel cell can be described by Eq. (19) [35].

$$P_{FC} = N_S \times V_{FC} \times I_{FC} \quad (19)$$

Furthermore, output voltage can be expressed as Eq. (20):

$$V_{FC} = E - V_{act} - V_{con} - V_{\Omega} \quad (20)$$

A PEM fuel cell uses an electrochemical reaction to convert chemical energy into electrical energy. Using hydrogen as an external fuel, the energy released during oxidation is converted into electricity, heat, and water, without harmful emissions. Combining fuel cells with HRESs enhances system sustainability by allowing electricity generation when renewable energy availability is low.

2.3.8. Grid integration

During power shortages, the grid provides the necessary energy. HOMER computes the total annual energy cost using Eq. (21) [33].

$$C_{AEC} = \sum_x \sum_y^{rates} E_{gp,x,y} P_{power,x} - \sum_x \sum_y^{rates} E_{gs,x,y} P_{sellback,x} \quad (21)$$

Eq. (22) used to determine the total yearly grid demand charge.

$$C_{gd} = \sum_x \sum_y^{rates} P_{pgd,x,y} D_x \quad (22)$$

2.4. Specifications of the system components

The assessment, shown in Table 1, includes all component costs such as installation, replacement, operation, and maintenance.

2.5. Economic modelling

2.5.1. Levelized cost of energy

HOMER calculates the LCOE using Eq. (23) [33].

Table 1

Economic and technical specifications for the various components used for the modeling [30].

Component	Parameters	Values
PV (Generic Flat Plate PV)	Rated capacity	1 kW
	Lifetime	25 years
	Capital cost	\$650
	Replacement cost	\$500
	Operation and maintenance cost	\$10/year
Wind turbine (Generic 3 kW)	Rated capacity	3 kW
	Rotor Diameter	15.81 m
	Lifetime	20 years
	Capital cost	\$3900
	Replacement cost	\$3500
Fuel cell (PEM)	Rated capacity	1 kW
	Initial/Capital cost	\$500
	Replacement cost	\$500
	Operation and maintenance cost	\$0.030/Op.Hour
	Lifetime (Hours)	15,000
Battery (EnerSys PowerSafe SBS 1800)	Nominal voltage	12 V
	Nominal capacity	24.8 kWh
	Roundtrip efficiency	97%
	Lifetime	15 years
	Capital cost	\$6500
Converter (system converter)	Replacement cost	\$6000
	Operation and maintenance cost	\$30.00/year
	Capacity	1 kW
	Efficiency (Inverter input)	95%
	Efficiency (Rectifier input)	95%
Grid	Lifetime	15 years
	Capital cost	\$460
	Replacement cost	\$460
	Operation and maintenance cost	\$10/year
	Grid power Price	Off-Peak = \$0.0950/kWh Peak = \$0.1320/kWh
Hydrogen tank	Sellback Price	Off-peak = \$0.05/kWh Peak = \$0.09/kWh
	Demand rate	\$0.90/kW/month
	Size	1 kg
	Capital cost	\$100
	Replacement cost	\$100
Electrolyzer (Generic)	Operation and maintenance cost	\$1
	Capacity	1 kW
	Capital cost	\$500
	Replacement cost	\$400
	Operation and maintenance cost	\$20
	Lifetime	15 years
	Efficiency	85%

$$LCOE = \frac{C_C + C_R + C_{OM} + C_f - S}{E_Y} \quad (23)$$

2.5.2. Net present cost

The NPC is calculated using Eq. (24), (25) and (26), where $CRF(p, q)$ denotes the capital recovery factor [33].

$$NPC = \frac{C_C + C_R + C_{OM} + C_f - S}{CRF(p, q)} \quad (24)$$

$$CRF(p, q) = \frac{p(1+p)^N}{(1+p)^N - 1} \quad (25)$$

$$p = \frac{p' - s}{p' + s} \quad (26)$$

2.5.3. Levelized cost of hydrogen

The LCOH is a critical parameter in evaluating the economic feasibility of a hydrogen production system. The LCOH can be calculated using Eq. (27):

$$LCOH = \frac{C_h + C_e}{\sum_{i=1}^T H_i} \quad (27)$$

The cost of electricity for hydrogen production (C_e) can be determined using Eq. (28):

$$C_e = LCOE \cdot \frac{\sum_{i=1}^T E_i}{T} \quad (28)$$

The total cost of electrolysis (C_h) is computed using Eq. (29):

$$C_h = \frac{C_{el} + P_e}{8760 \cdot T \cdot CF \cdot \eta} \quad (29)$$

2.6. Performance metrics for energy modeling

2.6.1. Capacity shortage fraction

The capacity shortage fraction indicates how much the system's capacity fails to meet the total demand. It is expressed as the ratio of the unmet demand to the total demand during a specific period. For HOMER's requirements, this fraction must not exceed the maximum annual capacity shortage. Eq. (30) is used to determine this metric [33].

$$f = \frac{E_T}{E_D} \quad (30)$$

2.6.2. Unmet load fraction

The unmet load fraction refers to the proportion of the total annual electrical demand that is not met due to insufficient power generation. It is computed using Eq. (31) [33].

$$F_{unmet} = \frac{E_U}{E_D} \quad (31)$$

2.6.3. Excess electricity fraction

The excess electricity fraction represents the proportion of electricity that exceeds the demand relative to the total electricity generated. HOMER computes this value at the end of each simulation using Eq. (32) [33].

$$F_E = \frac{E_{ex}}{E_p} \quad (32)$$

2.6.4. Renewable fraction

The renewable fraction indicates the percentage of a system's total energy output that is sourced from HRESs. This metric is vital for understanding the environmental benefits and long-term sustainability of

the system. It plays a critical role in evaluating the ecological suitability of systems modeled with HOMER. It is computed using Eq. (33) [17].

$$RF = \left(1 - \frac{\sum E_c}{\sum E_r}\right) \times 100 \quad (33)$$

2.7. Social criteria assessment

2.7.1. Human development index

The United Nations developed a composite index called the Human Development Index (HDI) to evaluate and compare the quality of life and well-being of populations across different nations. Unlike traditional economic indicators, HDI considers a broader range of human development factors, providing a more comprehensive picture of a country's progress. It is also an important measure for gauging the broader, often indirect, advantages brought about by various systems. A rise in HDI reflects enhancements in societal living standards, as illustrated by Eq. (34) [36].

$$HDI = 0.0978 \times \ln \left(\frac{E_{Load} + \min(F_{ex} \times E_{ex}, F_{ld} \times E_{Load})}{N_p} \right) - 0.0319 \quad (34)$$

2.7.2. Job potential

Since workers are required from the construction phase through the operation and decommissioning phase, energy system deployment typically results in the creation of jobs. Therefore, the study took into account the social aspects of the different energy systems to evaluate their potential to create jobs in the study area and beyond. Eq. (35) was used to determine the estimated job potentials for each of the different HRES components [17,37]:

$$Job = \sum_{x \in comp} job_x \cdot E_{gen,x} \quad (35)$$

2.8. Multicriteria decision-making approach

The selection of the most suitable HRES for electricity production and hydrogen generation is a complex process, particularly when multiple criteria related to technical, economic, environmental, and social factors are involved. In this study, the evaluation includes a comprehensive set of 10 sub-criteria spanning economic, technical, environmental, and social categories. Economic criteria include NPC, LCOE and LCOH. Technical factors comprise electricity production, hydrogen generation, unmet load, excess electricity, and renewable fraction. Environmental aspects include CO₂ emissions, and renewable energy utilization. Social criteria focus on job creation and the HDI. The CRITIC method was used to determine the weights of the criteria by analyzing their variability and correlation. TOPSIS was then employed to rank the HRES. Fig. 6 illustrates the flowchart of MCDM approaches (CRITIC and TOPSIS).

2.8.1. Criteria Importance Through Intercriteria Correlation (CRITIC) weight method

The CRITIC method determines objective weights based on the variability of criteria values and the conflict among criteria. The key characteristic of the CRITIC method is its ability to capture the intensity of constant variability and correlation between criteria [38]. The process for calculating weights is detailed as follows.

Step 1: Formulation of decision matrix

The decision matrix (X) is formulated to represent the performance values of m alternatives across n criteria, as shown in Eq. (36):

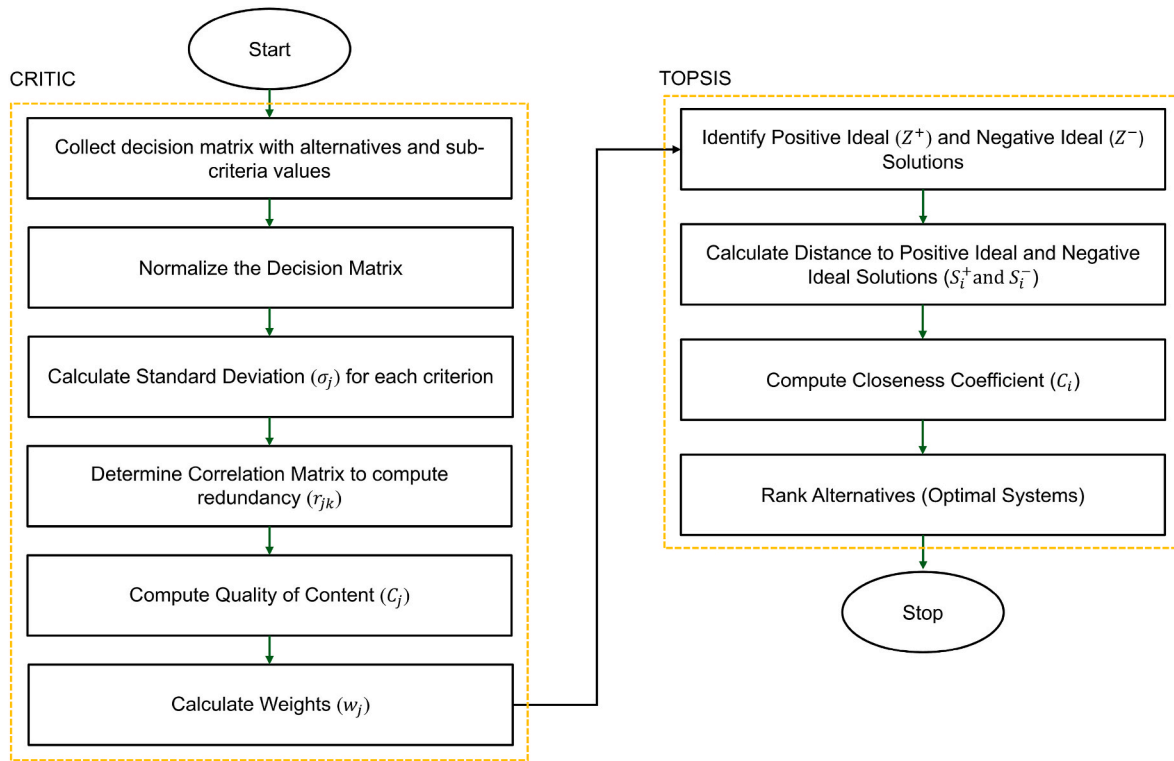


Fig. 6. Flowchart of MCDM approaches (CRITIC and TOPSIS) [Compiled by the authors].

$$X = x_{ij} = \begin{bmatrix} x_{11} & x_{12} & \cdots & x_{1n} \\ x_{21} & x_{22} & \cdots & x_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ x_{m1} & x_{m2} & \cdots & x_{mn} \end{bmatrix} \text{ where } i = 1, 2, \dots, m \text{ and } j = 1, 2, \dots, n \quad (36)$$

Step 2: Normalize the decision matrix

A normalized decision matrix is derived using Eq. (37):

$$r_{ij} = \frac{x_{ij} - x_j^{\min}}{x_j^{\max} - x_j^{\min}} \quad (37)$$

Step 3: Calculate standard deviation

The standard deviation (σ_j) of each criterion j is calculated using Eq. (38):

$$\sigma_j = \sqrt{\frac{1}{m} \sum_{i=1}^m (r_{ij} - \bar{r}_j)^2} \quad (38)$$

The standard deviation measures the variability or contrast intensity of a criterion.

Step 4: Calculate correlation coefficients

A symmetric $n \times n$ matrix of correlation coefficients (r_{jk}) is computed for all pairs of criteria using Eq. (39):

$$r_{jk} = \frac{\sum_{i=1}^m (r_{ij} - \bar{r}_j)(r_{ik} - \bar{r}_k)}{\sqrt{\sum_{i=1}^m (r_{ij} - \bar{r}_j)^2 \cdot \sum_{i=1}^m (r_{ik} - \bar{r}_k)^2}}; k = 1, 2, \dots, N \quad (39)$$

Step 5: Calculate the measure of conflict

The measure of conflict for each criterion j is determined as shown in Eq. (40):

$$\text{Conflict}_j = \sum_{k=1}^n (1 - r_{jk}) \quad (40)$$

This step quantifies the level of conflict created by a criterion with respect to others.

Step 6: Calculate the quality of information

The quality of information (C_j) represents the overall importance of a criterion, considering both its variability and independence. The quality of information for each criterion j is obtained by multiplying the standard deviation by the measure of conflict as shown in Eq. (41):

$$C_j = \sigma_j \cdot \sum_{k=1}^n (1 - r_{jk}) \quad (41)$$

Step 7: Compute objective weights

The final weights for each criterion (w_j) are calculated by normalizing the information content as shown in Eq. (42):

$$w_j = \frac{C_j}{\sum_{j=1}^n C_j} \quad (42)$$

Through the completion of the outlined process, the objective weights for all selection criteria are calculated. These weights are deemed appropriate provided they remain within the normalized range between 0 and 1.

2.8.2. Technique for Order Preference by similarity to ideal solution (TOPSIS)

In MCDM, attributes are categorized as either benefit or cost criteria to evaluate and compare alternative solutions. The decision matrix (X) is normalized to standardize all attributes into dimensionless units, enabling meaningful comparisons across different criteria [39]. The normalized decision matrix (R) is represented as Eq. (43):

$$R = \begin{bmatrix} r_{11} & r_{12} & \cdots & r_{1n} \\ r_{21} & r_{22} & \cdots & r_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ r_{m1} & r_{m2} & \cdots & r_{mn} \end{bmatrix} \quad (43)$$

Each element (r_{ij}) of the normalized matrix is calculated based on the type of attribute. The normalization is performed using Eq. (44):

$$r_{ij} = \frac{x_{ij}}{\sqrt{\sum_{i=1}^m x_{ij}^2}}, i = 1, 2, \dots, m; j = 1, 2, \dots, n \quad (44)$$

The weighted normalized decision matrix (V) is then calculated using the weights (w_j) (obtained from CRITIC) assigned to each criterion. The weights, w_j , are determined based on their relative importance and must satisfy the conditions $w_j \geq 0$ and $\sum_{j=1}^n w_j = 1$. The calculation of V is done using Eq. (45):

$$V = v_{ij} = r_{ij} \cdot w_j = \begin{bmatrix} v_{11} & v_{12} & \cdots & v_{1n} \\ v_{21} & v_{22} & \cdots & v_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ v_{m1} & v_{m2} & \cdots & v_{mn} \end{bmatrix} \quad (45)$$

Step 1: Determine the positive and negative ideal solutions

The ideal solution (Z^+) and the negative ideal solution (Z^-) are determined using Eq. (46) and (47):

$$z_j^+ = \begin{cases} \max(v_{ij}), & \text{if } f_j \in F_1 \text{ (Benefit objectives)} \\ \min(v_{ij}), & \text{if } f_j \in F_2 \text{ (Cost objectives)} \end{cases} \quad (46)$$

$$z_j^- = \begin{cases} \min(v_{ij}), & \text{if } f_j \in F_1 \text{ (Benefit objectives)} \\ \max(v_{ij}), & \text{if } f_j \in F_2 \text{ (Cost objectives)} \end{cases} \quad (47)$$

Step 2: Calculate the distance from positive and negative ideal solutions

The distances of each alternative from the positive ideal (Z^+) and negative ideal (Z^-) solutions are calculated using the Euclidean distance formulae Eq. (48) and Eq. (49):

$$S_i^+ = \sqrt{\sum_{j=1}^n (v_{ij} - z_j^+)^2}, i = 1, 2, \dots, m \quad (48)$$

$$S_i^- = \sqrt{\sum_{j=1}^n (v_{ij} - z_j^-)^2}, i = 1, 2, \dots, m \quad (49)$$

Step 3: Calculate the closeness coefficient

The closeness coefficient (G_i) of each alternative to the ideal solution is calculated using Eq. (50):

$$G_i = \frac{S_i^-}{S_i^+ + S_i^-}, i = 1, 2, \dots, m \quad (50)$$

Step 4: Rank Alternatives

Rank alternatives based on the closeness coefficient G_i , where a higher G_i indicates a better alternative as shown in Eq. (51).

$$\text{Rank}_i = \text{Rank}(G_i, \text{descending order}) \quad (51)$$

3. Result and discussion

This section provides a comprehensive overview of the optimization results, including detailed cost analysis, electrical performance metrics, emission impact assessment, and sensitivity analysis. Each component is explored to offer clear insights into system efficiency and economic viability.

3.1. Optimization result

The grid-tied systems in this study present varied performance outcomes, particularly in terms of cost metrics and renewable energy utilization. Table 2 summarizes optimized results, which include LCOE, LCOH, NPC, resource use, and electricity production. All systems follow the LF dispatch strategy after optimization. The configurations reveal distinct component capacities. HRES4 has the highest PV capacity at 7036 kW, followed by HRES1 at 6618 kW, HRES3 at 6552 kW and HRES2 at 5489 kW. Wind energy contributes 1500 kW to HRES1 and HRES3, while HRES2 and HRES4 rely solely on solar power. Fuel cells, at 1600 kW, are present in HRES1 and HRES2 in hydrogen storage; HRES3 and HRES4 do not include fuel cells. Energy storage differs as well, with HRES1 using 3 strings, HRES4 using 1, and HRES2 and HRES3 omitting storage. Each system includes electrolyzers rated at 1500 kW, but HRES2's hydrogen tank, with a capacity of 700 kg, exceeds the 500 kg capacity in HRES1, HRES3, and HRES4, allowing more extended hydrogen storage. Converter capacities also vary: HRES3 at 2551 kW, HRES4 at 2332 kW, HRES2 at 1013 kW, and HRES1 at 913 kW, showing different levels of AC/DC conversion capabilities.

The economic analysis shows clear distinctions among the systems. HRES3, with the lowest LCOE at \$0.0321/kWh, ranks as the most efficient configuration, while HRES2, with the highest LCOE at \$0.1100/kWh, ranks lowest in efficiency. For LCOH, HRES1 and HRES3 stand close at \$5.17/kg and \$5.22/kg, making them the most economical choices. In contrast, HRES2 and HRES4 have higher LCOH values, with HRES2 reaching \$7.76/kg. Regarding NPC, HRES3 proves the most cost-effective at \$10.11 million, while HRES2 has the highest NPC at \$13.57 million, showing a greater lifetime cost. However, HRES3 leads with a renewable fraction of 91.7%, reflecting strong reliance on HRESs.

Table 2
Optimization results of HRES configurations.

Parameter	HRES1	HRES2	HRES3	HRES4
	PV-Wind-Battery-Fuel cell-Grid	PV-Battery-Fuel cell-Grid	PV-Wind-Battery-Grid	PV-Battery-Grid
PV (kW)	6618	5489	6552	7036
Wind energy (kW)	4500	–	4500	–
Fuel cell (kW)	1600	1600	–	–
Battery storage (strings)	3	–	–	1
Grid (kW)	999,999	999,999	999,999	999,999
Electrolyzer (kW)	1500	1500	1500	1500
Hydrogen tank (kg)	500	700	500	500
Converter (kW)	913	1013	2551	2332
Dispatch	LF	LF	LF	LF
LCOE (\$/kWh)	0.04716	0.1100	0.0321	0.0777
LCOH (\$/kg)	5.17	7.76	5.22	7.20
NPC (\$)	11.98 M	13.57 M	10.11 M	13.25 M
Renewable fraction (%)	89.7	53.2	91.7	67.0

HRES1 follows with an 89.7% renewable fraction, demonstrating a balanced mix of solar and wind. HRES4, with 67.0%, and HRES2, at 53.2%, depend more on grid and non-RES. Consequently, HRES3 emerges as the system with the highest renewables, while HRES2 has the lowest. Overall, HRES3 not only excels in cost-effectiveness for both electricity and hydrogen but also has the highest renewable reliance, that presents itself as the most balanced and sustainable option among the designs.

3.2. Production and consumption dynamics of electricity and hydrogen

The electrical summary in Table 3 outlines key differences among the designed systems in terms of energy production, consumption, and grid reliance. HRES1 and HRES3 produce the most electricity annually, generating 18,387,085 kWh/yr and 18,079,979 kWh/yr, respectively. In contrast, HRES2 and HRES4 generate significantly less, with 9,858,432 kWh/yr and 11,880,342 kWh/yr. HRES3 leads in consumption with an annual demand of 17,230,100 kWh, followed by HRES1 at 15,394,636 kWh. HRES4 and HRES2 have lower consumption rates at 10,899,481 kWh/yr and 8,708,406 kWh/yr, which align with their reduced production levels. All systems purchase grid energy to supplement their generation, but HRES2 and HRES4 rely more heavily on the grid, purchasing 2,457,753 kWh/yr and 2,394,884 kWh/yr, respectively. HRES1 and HRES3 purchase only around 1,118,110 kWh each. Regarding grid sales, HRES3 stands out with 8,793,239 kWh sold back, demonstrating effective utilization and surplus generation. HRES1 also contributes significantly, selling 6,204,799 kWh, while HRES4 and HRES2 sell 2,652,767 kWh and 643,403 kWh, respectively. Excess electricity generation further highlights the efficiency and grid interaction of each system. HRES1 leads with an excess of 2,803,819 kWh/yr (15.2% of its production), showcasing strong energy productivity and surplus potential for grid sales. HRES2 follows with 1,002,890 kWh/yr (10.2%) in excess, demonstrating efficient generation. HRES4 generates 724,659 kWh/yr in excess (6.1%), indicating room for optimization. Despite its high production, HRES3 has the lowest excess at 551,842 kWh/yr (3.05%), suggesting it operates near capacity with minimal

Table 3
Comparative performance metrics of HRES configurations for electricity and hydrogen production.

Parameter	HRES1 PV-Wind- Battery-Fuel cell-Grid	HRES2 PV-Battery- Fuel cell- Grid	HRES3 PV-Wind- Battery- Grid	HRES4 PV- Battery- Grid
Annual electricity production (kWh/yr)	18,387,085	9,858,432	18,079,979	11,880,342
Annual electricity consumption (kWh/yr)	15,394,636	8,708,406	17,230,100	10,899,481
Annual grid energy purchased (kWh)	1,118,110	2,457,753	1,118,234	2,394,884
Annual grid energy sold (kWh)	6,204,799	643,403	8,793,239	2,652,767
Excess electricity (kWh/yr)	2,803,819 (15.2%)	1,002,890 (10.2%)	551,842 (3.05%)	724,659 (6.10%)
Annual hydrogen production (kg/yr)	98,693	74,329	82,467	78,418
Annual hydrogen consumption (kg/yr)	98,315	73,845	82,043	77,997

surplus for grid sales. All systems meet energy demands consistently, with zero unmet electric load and no capacity shortages. HRES1 and HRES3 rank as the most productive systems, and HRES3 achieves high grid sales with limited excess generation. HRES2 and HRES4 perform efficiently but rely more on grid purchases and generate less excess electricity.

Fig. 7 shows the monthly electricity generation for each system. For HRES1 and HRES3, wind energy makes the largest contribution that peaks during the summer months (June to August) and declines from September onward. PV and grid contributions remain consistent throughout the year. In contrast, HRES2 and HRES4 rely solely on PV and grid, resulting in lower total production. PV energy dominates the generation, particularly during the summer months (March to August), while the grid consistently provides a smaller, steady backup.

Fig. 8 presents the monthly hydrogen production from each system. Both HRES1 and HRES3 maintain stable hydrogen production throughout the year, with levels remaining high and only slight fluctuations. Production remains between 200 and 250 kg/year, without significant seasonal variation, demonstrating optimization for consistent hydrogen generation. In contrast, HRES2 and HRES4 show more variability in hydrogen output. HRES2 peaks in production in April, dips from July to September, and recovers in the final months of the year. HRES4 follows a similar trend, though its dip in production from July to September is less pronounced. Production for both systems ranges between 150 and 250 kg/year, showing sensitivity to seasonal factors, such as solar or wind availability. In summary, HRES1 and HRES3 may better suit applications requiring consistent, year-round hydrogen generation, while HRES2 and HRES4 experience seasonal influences that affect hydrogen output in certain months.

The hydrogen tank levels over a year for the four HRESs are displayed in Fig. 9. HRES1 and HRES3 maintain high hydrogen levels, with storage generally ranging from 375 to 500 kg throughout the year. This stability indicates a reliable energy supply, likely due to an optimized balance between HRESs and storage, that ensures steady hydrogen production without significant depletion. In contrast, systems HRES2 and HRES4 show greater variability, with hydrogen levels frequently dropping, especially during mid-year and summer months. In summary, HRES1 and HRES3 maintain more stable hydrogen levels, while HRES2 and HRES4 face greater challenges due to energy supply variability.

3.3. Economic analysis

The cost summaries, grouped by cost type, of the four HRESs shown in Table 4. HRES1 requires the highest initial investment of \$12.22 million but has operational benefits with a negative O&M cost of -\$2.45 million, which indicates potential revenue to offset expenses. HRES3 stands as the most cost-effective option with the lowest NPC of \$10.1 million, aided by efficient operations and a positive salvage value of \$6.19 million, despite a high initial capital of \$12.09 million. In contrast, HRES2, with the lowest initial capital of \$5.65 million, incurs the highest O&M costs at \$7.27 million, resulting in the highest NPC of \$13.57 million and a less favorable financial outlook. HRES4 has an NPC of \$13.25 million, warranting further review of its operational efficiency. Overall, HRES3 emerges as the best choice, while HRES2's high O&M costs raise concerns about its long-term viability despite lower initial costs.

Fig. 10 shows the cost distribution for components across each system. HRES1 and HRES3 rely heavily on the grid, with negative GRID costs of -\$5.6 million and -\$8.6 million, respectively, raising their operational expenses. In contrast, HRES2 and HRES4 show positive grid costs of \$5 million and \$1.9 million, indicating either surplus energy generation for sale or reduced grid reliance. For PV investment, HRES4 leads with \$7 million, emphasizing solar energy. HRES1 and HRES3 invest \$6.6 million and \$6.5 million, respectively, but offset with greater grid dependence. HRES2, with \$5.5 million in PV, maintains a balanced approach. HRES3 and HRES4 have the highest converter costs at \$2.5

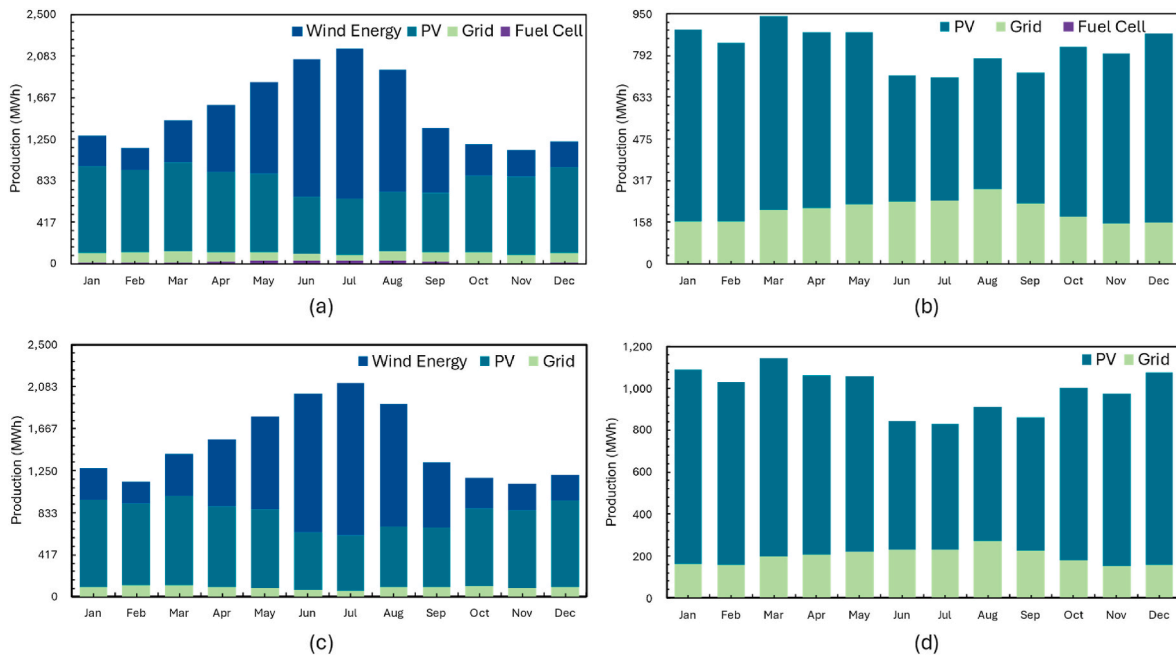


Fig. 7. Monthly electricity production for (a) HRES1 (PV-Wind-Battery-Fuel cell-Grid) (b) HRES2 (PV-Battery-Fuel cell-Grid) (c) HRES3 (PV-Wind-Battery-Grid) (d) HRES4 (PV-Battery-Grid).

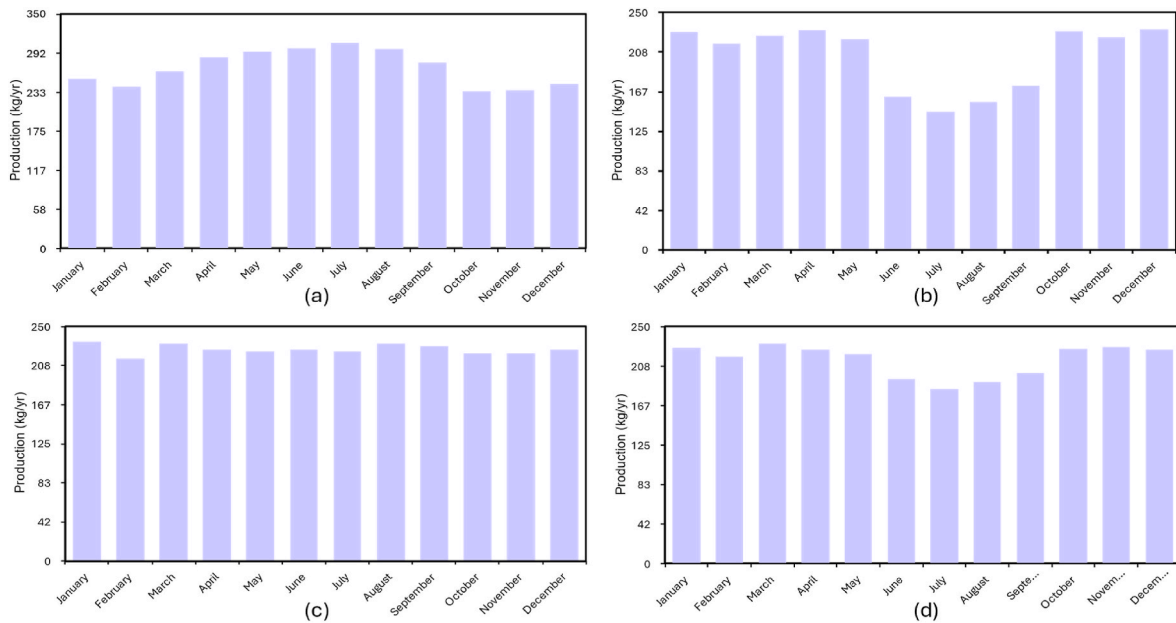


Fig. 8. Monthly Hydrogen production from electrolyzer for (a) HRES1 (PV-Wind-Battery-Fuel cell-Grid) (b) HRES2 (PV-Battery-Fuel cell-Grid) (c) HRES3 (PV-Wind-Battery-Grid) (d) HRES4 (PV-Battery-Grid).

million and \$2.3 million, suggesting advanced conversion requirements. Wind turbine costs reach \$7.6 million for HRES2 and HRES3, focusing on wind energy. HRES4 has minimal battery costs at \$22,026, compared to HRES1's \$66,078, indicating a smaller or more advanced storage need. Overall, HRES4 prioritizes solar energy and minimal grid reliance, HRES1 and HRES3 depend more on the grid despite high renewable investments, and HRES2 balances renewable and grid reliance.

3.4. Emission analysis

The emission comparison of the four systems is shown in Table 5. HRES1 and HRES3 have lower CO₂ emissions at 706,223 kg/yr and

706,724 kg/yr, respectively, while HRES2 and HRES4 have much higher emissions at 1,553,300 kg/yr and 1,513,567 kg/yr. This demonstrates the benefits of wind turbines, which significantly reduce CO₂ output. In terms of CO, HRES1 is the only system with measurable emissions (268 kg/yr), while HRES2, HRES3, and HRES4 produce none. Unburned hydrocarbons appear only in HRES1 at 11.7 kg/yr, further indicating inefficiencies in its fuel conversion process, while HRES2, HRES3, and HRES4 show zero emissions. Additionally, HRES1 produces 1.63 kg/yr of particulate matter (PM), while the other systems produce none, suggesting that HRES1's processes contribute to particulate emissions. HRES1 and HRES3 have SO₂ emissions of 3064 kg/yr, while HRES2 and HRES4 have higher emissions at 6734 kg/yr and 6562 kg/yr,

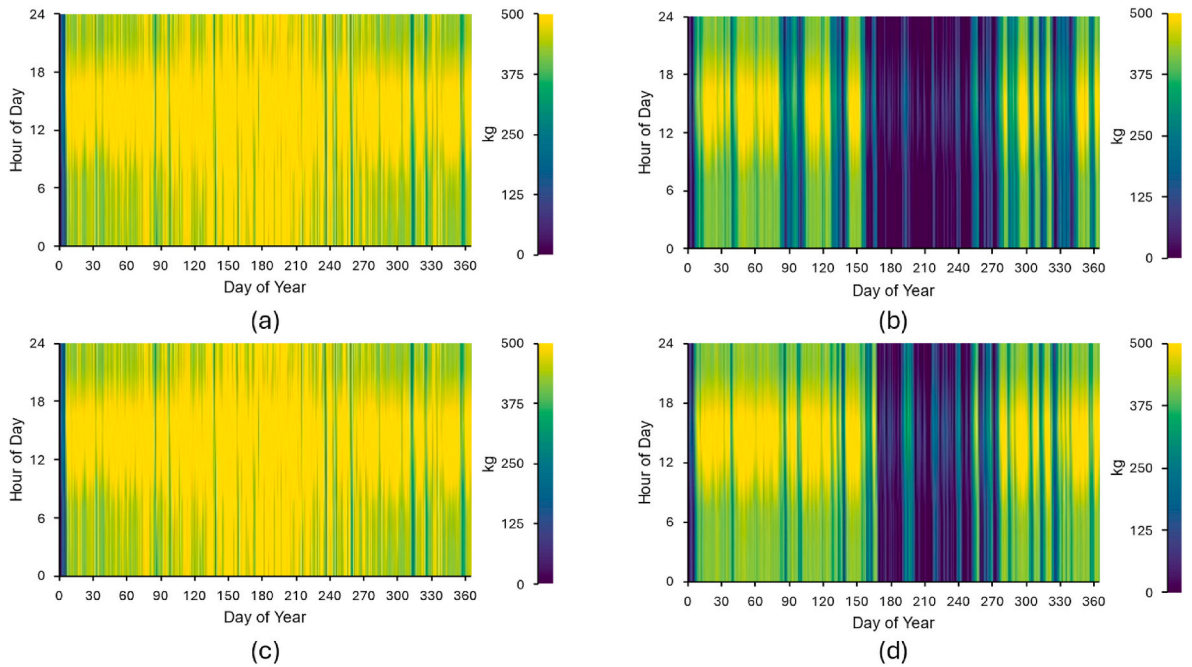


Fig. 9. Hydrogen tank level for (a) HRES1 (PV-Wind-Battery-Fuel cell-Grid) (b) HRES2 (PV-Battery-Fuel cell-Grid) (c) HRES3 (PV-Wind-Battery-Grid) (d) HRES4 (PV-Battery-Grid).

Table 4
Cost summary by cost type.

Parameter	HRES1	HRES2	HRES3	HRES4
	PV-Wind-Battery-Fuel cell-Grid	PV-Battery-Fuel cell-Grid	PV-Wind-Battery-Grid	PV-Battery-Grid
Initial capital (\$)	12,221,365.57	5,653,985.78	12,093,176.46	6,469,619.69
O & M cost (\$/yr)	−2,451,216.28	7,267,274.02	−84,553	288,648
Salvage value (\$)	−6,540,415.99	−2,830,785.16	6,191,670.27	2,836,208.01
Replacement cost (\$)	8,753,586.91	3,483,584.48	9,391,169.56	4,761,110.40
Total NPC (\$)	11,983,320.21	13,574,059.11	10,106,700.38	13,251,075.08



Fig. 10. Cost summary by component type.

respectively. This suggests that the absence of wind turbines may limit the systems' ability to reduce SO₂ emissions linked to fossil fuel combustion. Finally, NO_x emissions are lowest in HRES3 (1498 kg/yr), with HRES1, HRES2, and HRES4 showing higher outputs of 1750 kg/yr, 3293 kg/yr, and 3209 kg/yr, respectively. This highlights the positive effect of integrating wind energy. Overall, HRES1 and HRES3 are more environmentally friendly and support a cleaner energy future.

Table 5
Emissions of the designed systems.

Parameter	HRES1	HRES2	HRES3	HRES4
	PV-Wind-Battery-Fuel cell-Grid	PV-Battery-Fuel cell-Grid	PV-Wind-Battery-Grid	PV-Battery-Grid
Carbon Dioxide (kg/yr)	706,223	1,553,300	706,724	1,513,567
Carbon Monoxide (kg/yr)	268	0	0	0
Unburned hydrocarbons (kg/yr)	11.7	0	0	0
Particulate matter (kg/yr)	1.63	0	0	0
Sulfur Dioxide (kg/yr)	3064	6734	3064	6562
Nitrogen oxides (kg/yr)	1750	3293	1498	3209

3.5. Social criteria assessment

Fig. 11 shows the job potential (persons in 25 years) and Human development index for designed systems.

3.5.1. Job potential analysis

Deployment of HRES not only meets energy demands but also

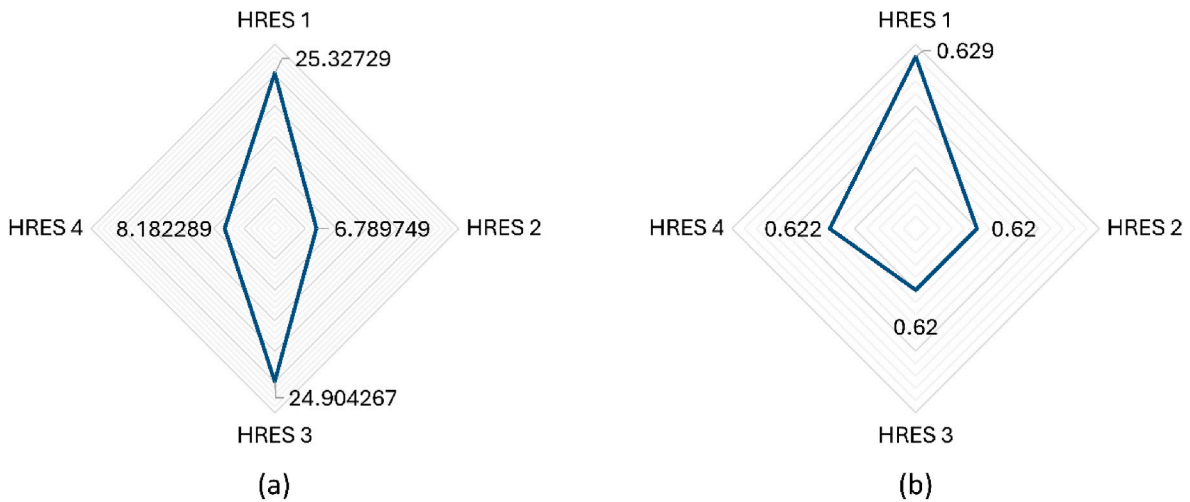


Fig. 11. (a) Job potential (b) Human development index for designed systems.

provides socioeconomic benefits in the form of job creation. The analysis considers energy generation by individual renewable energy components such as solar PV, wind energy, and micro hydropower. HRES1 and HRES3 exhibit the highest job potential, as seen in Fig. 11 (a), with scores of 25.33 and 24.90 people in 25 years, respectively. These HRESs also benefit from higher electricity production, which translates into greater employment opportunities across construction, operation, and maintenance phases. HRES4 shows a much lower job potential (8.18), which reflects its reliance on fewer renewable energy components and lower electricity production compared to HRES1 and HRES3. HRES2, despite having a solar PV component, records the lowest job potential (6.79), likely due to its reduced energy output and lack of wind energy integration.

3.5.2. Human development index analysis

From Fig. 11 (b), it can be observed that HRES1, with the highest HDI value of 0.629, represents the most balanced and optimal configuration for human development. This is primarily due to its integration of diverse renewable energy sources and storage systems, which ensure reliability and resilience in meeting energy demands. HRES2, with a slightly lower HDI of 0.622, shows a modest decrease due to the removal of the wind turbine. However, the HDI remains high, indicating that the system can still adequately support human development, albeit with reduced energy diversity and resilience compared to HRES1. HRES3, with an HDI value of 0.620, includes a wind turbine but lacks a fuel cell, which reduces the system's robustness by compromising energy security and reliability, especially during periods of high demand or low renewable energy production. Although HRES3 integrates renewable energy systems effectively, its reduced capacity for consistent energy supply may limit its ability to meet the needs of regions with variable weather patterns or fluctuating consumption levels. HRES4, also with an

HDI value of 0.620, omits both the wind turbine and the fuel cell, further diminishing the system's energy security and reliability. This configuration heavily relies on fewer energy sources and lacks sufficient backup power and storage, limiting its ability to provide a stable energy supply during high demand or adverse weather conditions. While HRES4 is functional and contributes to human development, it may not adequately address long-term energy needs or provide the level of reliability and sustainability required to fully support human development objectives.

3.6. Multicriteria decision-making analysis

The results from the CRITIC analysis from Table 6 highlight the relative importance of various criteria and sub-criteria in evaluating HRES. Each criterion's importance was determined based on its standard deviation (σ_j), information content (C_j), and calculated weight (w_j). The weights represent the proportionate significance of each criterion in percentage terms. The job potential sub-criterion under the societal category emerged as the most influential factor, with a weight of 25.67%. This indicates that the potential for job creation is the most critical consideration in the current evaluation, reflecting its significant socio-economic impact. Following this, excess electricity (%) under the technical category ranked second with a weight of 16.93%, signifying its importance in ensuring system efficiency and minimizing energy wastage. In the societal category, HDI also held substantial importance, with a weight of 10.22%. This emphasizes the role of societal benefits, such as improved quality of life and equitable development, in the assessment process. Among the environmental sub-criteria, CO₂ emissions had the highest weight of 7.55%, indicating the criticality of reducing carbon footprints in hybrid energy systems. From the economic perspective, NPC and LCOE stood out with weights of 6.88% and 5.87%,

Table 6

Standard deviation, Information content and Relative weight of importance of 10 sub-criteria according to CRITIC.

Criteria	Sub-criteria	σ_j	C_j	w_j	w_j (%)
Economic	LCOE	0.44313	1.955575	0.058658	5.865830489
	NPC	0.453517	2.294536	0.068826	6.882559539
	LCOH	0.517013	2.239047	0.067161	6.716118441
Technical	Excess electricity (%)	0.461268	5.643204	0.16927	16.92703419
	Electricity production	0.508039	2.196032	0.065871	6.587093829
	Hydrogen production	0.43812	2.450989	0.073518	7.351847518
Environment	CO ₂ emission	0.563964	2.518438	0.075542	7.554163595
	Renewable fraction	0.482115	2.073512	0.062196	6.219589705
Society	HDI	0.474667	3.408359	0.102235	10.22351947
	Job potential	0.550012	8.558717	0.256722	25.67224322

respectively. These metrics underline the need for cost-effective solutions in energy planning. Similarly, LCOH was also significant, with a weight of 6.72%. Technical sub-criteria such as electricity production and H₂ production were weighted at 6.59% and 7.35%, respectively, reflecting their roles in ensuring energy output and hydrogen utilization efficiency. Additionally, renewable fraction, under the environmental category, contributed a weight of 6.22%, emphasizing the importance of renewable energy integration in hybrid systems.

The TOPSIS analysis from Table 7 provides a comprehensive evaluation of HRES by ranking alternatives based on their proximity to the ideal solution (S_i^+) and their distance from the negative ideal solution (S_i^-). The closeness coefficient (C_i) is used as the primary metric to determine the rankings, with higher values indicating a stronger alignment with the ideal solution. Among the evaluated systems, HRES3 emerged as the most optimal choice with a C_i value of 0.556127, indicates that HRES3 is the closest to the ideal solution and demonstrates superior performance across the critical evaluation criteria. Its strong position suggests that it effectively balances economic, environmental, and technical parameters, making it the preferred system for implementation. HRES1 secured the second rank with a C_i value of 0.591756639. While slightly behind HRES3, HRES1 demonstrates significant potential as a competitive alternative. HRES4 achieved the third position with a C_i value of 0.441687512. Although its performance is commendable, it falls slightly behind HRES3 and HRES1 in terms of overall proximity to the ideal solution. HRES4 may still be considered a strong candidate, depending on the prioritization of specific criteria in decision-making. HRES2, with a C_i value of 0.336515462, ranked fourth and is the least favorable among the evaluated systems. Its lower ranking suggests that it may not perform as effectively in meeting the desired objectives compared to the other alternatives. This system could require further optimization or may be better suited for applications with less stringent requirements.

3.7. Sensitivity analysis

HOMER Pro sensitivity analysis is a robust method to evaluate how variations in input parameters influence the performance and cost-effectiveness of HRESs. By systematically adjusting key factors—such as resource availability, component costs, and operational constraints—HOMER Pro enables users to explore a variety of scenarios and determine which variables have the most substantial effect on system results.

3.7.1. Economic impact of hydrogen load variability

Fig. 12 provides insights into how varying levels of unmet hydrogen demand and scaled average loads impact the NPC and LCOE. Higher unmet hydrogen demand reduces NPC due to lower operational requirements but can increase LCOE due to inefficiencies. Conversely, lower unmet demand raises NPC due to additional investments in system components but reduces LCOE through efficient resource utilization. Increased loads generally raise NPC due to higher capacity requirements but may lower LCOE by leveraging economies of scale. For instance, in Fig. 12 (a), the NPC ranges from approximately \$6.75 M to \$8.97 M. The LCOE varies between 0.025 and 0.038. As the scaled average load increases from 150 kWh/d to 225 kWh/d, the NPC rises significantly. As hydrogen demand increases, systems with a high share of renewables and storage show lower unmet load percentages. For instance, at 225

kWh/d, the NPC difference between 5% and 20% unmet hydrogen is relatively small compared to the changes seen at 150 kWh/d. Fig. 12 (b) exhibits a higher NPC range, between \$10.8 M and \$13.3 M, with LCOE spanning from 0.063 to 0.106. The NPC and LCOE patterns indicate that lower unmet hydrogen percentages at higher loads contribute to a lower LCOE, while lower loads paired with higher unmet hydrogen percentages result in the highest costs. This suggests a more sensitive dependence of costs on unmet hydrogen percentages in this scenario. Fig. 12 (c) closely resembles Fig. 12 (a) in both NPC and LCOE ranges, which suggests similar system behavior under these configurations. However, slight deviations are observed, such as marginally higher costs for some load and unmet hydrogen combinations. This suggests that the underlying assumptions or external conditions might slightly differ from those in Fig. 12 (a). In Fig. 12 (d), the NPC ranges from \$12.0 M to \$13.8 M, with LCOE spanning from 0.053 to 0.098. Here, the trend reveals that reducing the unmet hydrogen percentage and increasing the scaled load minimizes LCOE. For example, at 5% unmet hydrogen and 225 kWh/d, LCOE is at its lowest in this subplot. Overall, the scaled average load has a stronger influence on the NPC than the Max Unmet hydrogen (%). However, LCOE is more sensitive to variations in unmet hydrogen, especially at lower loads. The variations across the subplots indicate that scenarios with higher loads and lower unmet hydrogen percentages are more cost-efficient. Critical analysis suggests that optimizing hydrogen supply to minimize unmet demand while maintaining higher loads can achieve lower costs and improve system efficiency.

3.7.2. Economic impact of CO₂ penalty and discount rate

A carbon tax (or carbon penalty) is added to encourage the reduction of emissions by placing a financial cost on carbon emissions which sends an economic signal to industries and investors, making sustainable, low-carbon technologies more attractive. By placing a financial cost on carbon emissions, it creates an economic signal that encourages the reduction of greenhouse gas emissions, aligning with global climate goals. This mechanism promotes innovation in renewable energy, energy efficiency, and carbon capture technologies, fostering a transition to cleaner energy systems. Furthermore, revenue generated from carbon taxes can be reinvested into green projects, infrastructure, or used to mitigate social and economic impacts, contributing to a just energy transition.

In Fig. 13 (a), the NPC values range from approximately \$0.7 M to \$1.1 M. The LCOE values remain consistent across CO₂ penalties, indicating a limited influence of CO₂ penalties on energy costs under these conditions. For instance, the LCOE at a 4% discount rate stays around 0.011, regardless of the CO₂ penalty. As the nominal discount rate increases, the LCOE rises, with values reaching 0.055 at a 12% discount rate. That means higher discount rates amplify the effect of financial costs on energy economics, even when the CO₂ penalty is unchanged. In contrast, Fig. 13 (b) shows the LCOE varies significantly, from 0.036 at low discount rates and penalties to 0.133 at the highest rates and penalties. This indicates that CO₂ penalties substantially impact the NPC and LCOE when financial discount rates are higher. For example, at an 8% discount rate, the LCOE increases gradually from 0.095 to 0.102 as the penalty increases. Fig. 13 (c) reveals a similar pattern, with NPC values ranging from \$1.2 M to \$1.8 M. The LCOE values exhibit notable variation across both discount rates and CO₂ penalties. At a 4% discount rate, the LCOE starts at 0.053 and climbs to 0.058 with increasing penalties. However, the LCOE at a 12% discount rate remains consistently high, from 0.107 to 0.113, suggesting diminishing returns in cost reductions as penalties increase. Fig. 13 (d) mirrors the behavior of Fig. 13 (a), the LCOE values show minor sensitivity to changes in CO₂ penalties, particularly at lower discount rates. For example, at a 4% discount rate, the LCOE only slightly increases from 0.011 to 0.012. At higher discount rates, such as 12%, the LCOE remains consistent at 0.055, reflecting limited impact from penalty variations.

NPC is consistently influenced by both factors; LCOE demonstrates nonlinear trends. Low discount rates buffer the impact of CO₂ penalties

Table 7

Overall ranks of HRES according to TOPSIS.

HRES	S_i^+	S_i^-	C_i	Rank _i
1	0.061552787	0.089221954	0.591756639	2
2	0.095307173	0.048339239	0.336515462	4
3	0.010978372	0.110046159	0.90928804	1
4	0.076664499	0.060650178	0.441687512	3

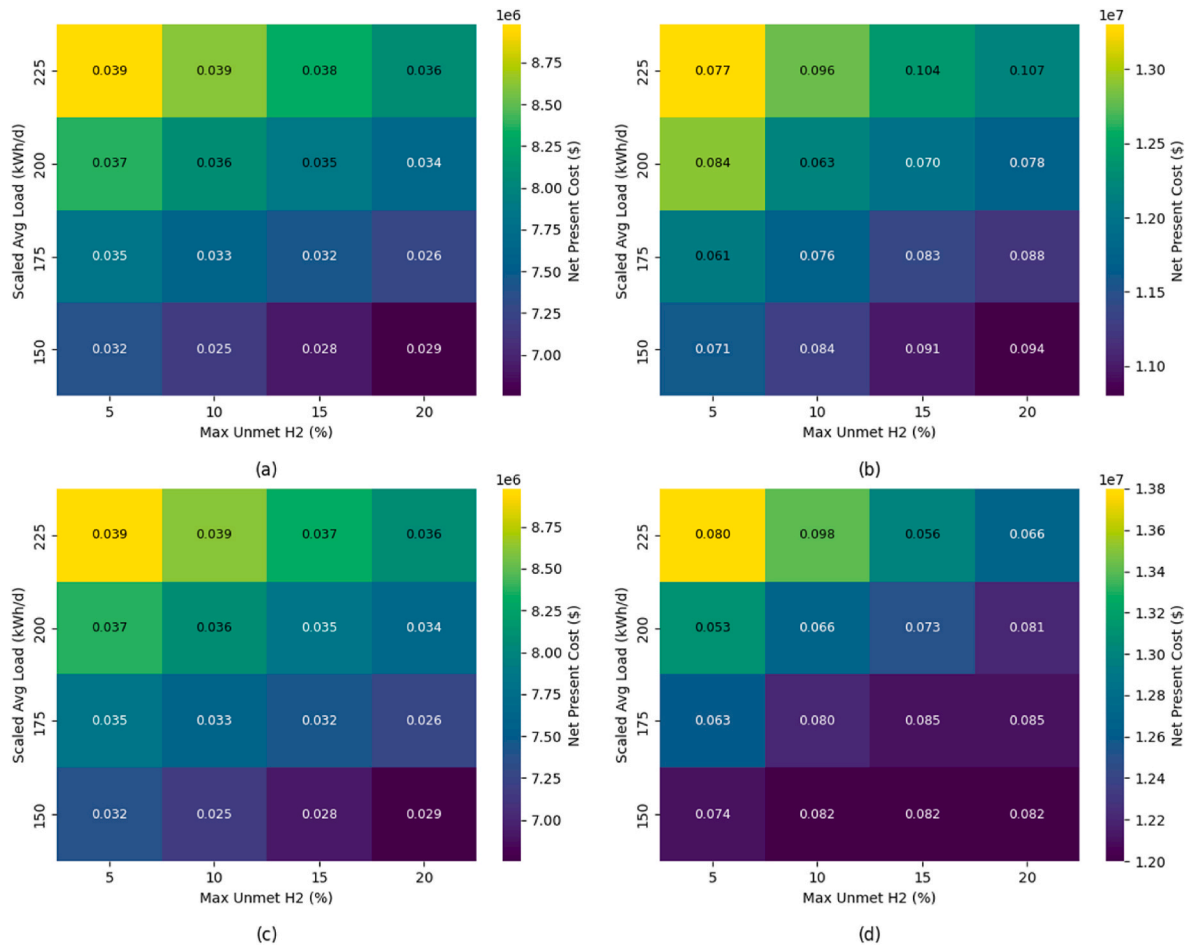


Fig. 12. Sensitivity analysis of hydrogen load dynamics and economic implications for (a) HRES1 (PV-Wind-Battery-Fuel cell-Grid) (b) HRES2 (PV-Battery-Fuel cell-Grid) (c) HRES3 (PV-Wind-Battery-Grid) (d) HRES4 (PV-Battery-Grid).

on costs, whereas high discount rates amplify their effects. The insights suggest that controlling financial discount rates is key to mitigating cost increases in scenarios with higher CO₂ penalties. Additionally, the diminishing LCOE variations in certain conditions indicate opportunities for optimizing system performance to balance financial and environmental costs effectively.

3.8. Comparative analysis with other studies

This study validates results by comparing green hydrogen production costs in Cox's Bazar, Bangladesh with those from studies conducted in other regions worldwide, as shown in Table 8. The comparison reveals that the costs in this study are comparable to or lower than those in previous studies, which shows hydrogen production in Bangladesh can be both affordable and economically feasible using the proposed HRESs in this study.

4. Conclusions

This study evaluates the techno-economic viability of grid-connected HRES for hydrogen production in Cox's Bazar, Bangladesh, a region with favorable climatic conditions for renewable energy generation. The research analyzes four distinct HRES configurations incorporating solar PV, wind turbines, hydrogen storage, and fuel cells. A multicriteria decision-making framework, combining CRITIC and TOPSIS, was used to rank the systems based on economic, technical, environmental, and social criteria, including job creation and the HDI. Sensitivity analysis was conducted to assess the impacts of hydrogen load variability, CO₂

penalties, and discount rates on the systems' economic and operational performance. From a social perspective, HRES1 and HRES3 generated the highest job creation potential, attributed to their integration of both solar PV and wind energy. Environmentally, these systems significantly reduced CO₂ emissions, underscoring the benefits of wind integration in hybrid configurations.

The findings reveal significant differences in the efficiency, cost-effectiveness, and environmental performance of the configurations. HRES3 emerged as the most optimal system through TOPSIS ranking, achieving the lowest LCOE (\$0.0321/kWh) and NPC (\$10.11 million) while maintaining the highest renewable fraction (91.7%) and an LCOH of \$5.22/kg. It also demonstrated stable hydrogen production year-round, minimal unmet loads, and the highest grid sales, making it the most resilient and financially viable configuration. HRES1, despite its higher initial capital investment, showed strong operational efficiency, while HRES2 and HRES4 exhibited higher costs, lower renewable fractions, and greater reliance on grid purchases. Environmentally, HRES1 and HRES3 significantly reduced CO₂ emissions compared to HRES2 and HRES4, utilizing the benefits of wind integration. The HDI analysis further confirmed the importance of diverse energy systems in enhancing human development, with HRES1 achieving the highest HDI value due to its reliability and sustainability.

The study's primary limitation lies in its reliance on the HOMER Pro optimizer, which, while widely used for modeling, simulating, and optimizing hybrid systems, has inherent constraints in capturing the full complexity of energy systems and advanced optimization techniques. The absence of robust metaheuristic methods in this model may limit its capacity to explore non-linear and multi-objective optimization spaces,

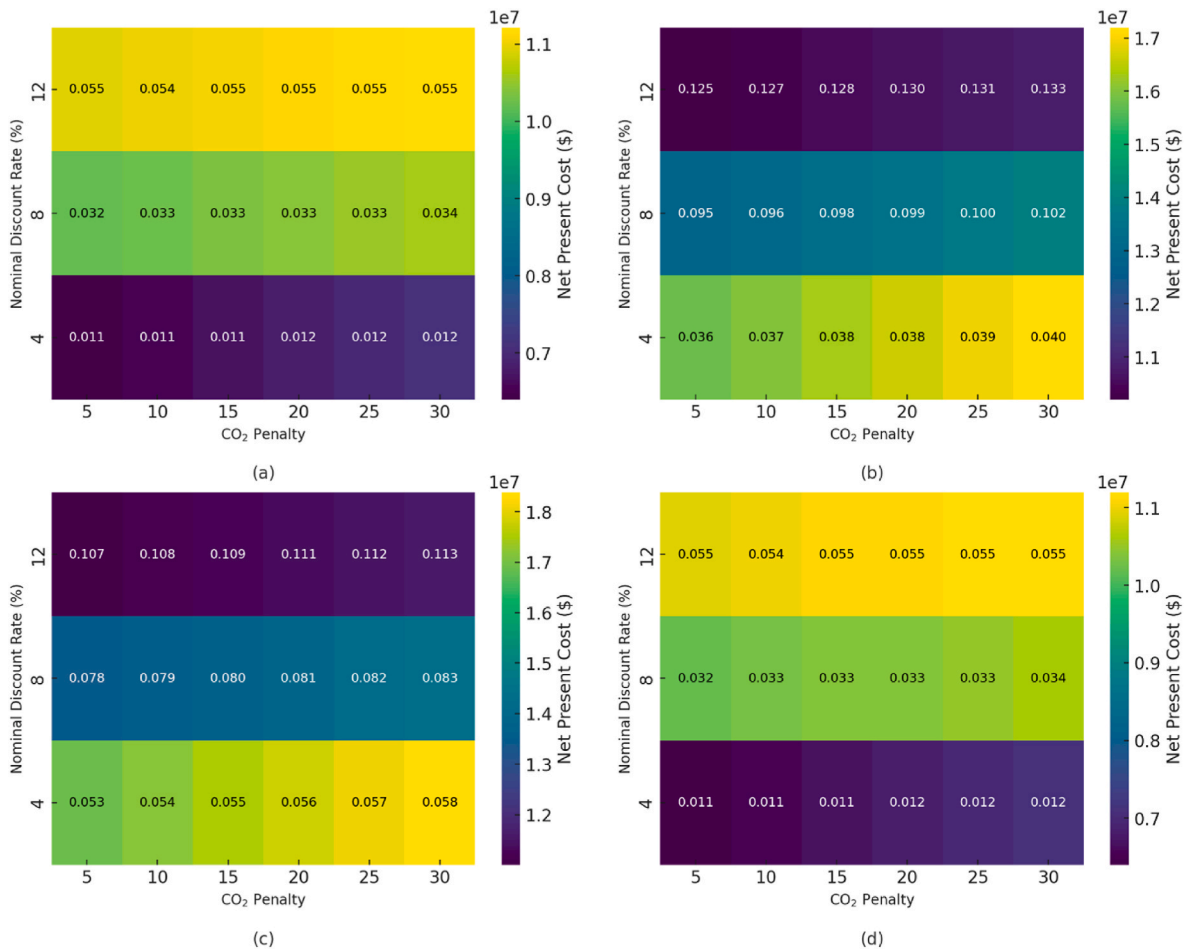


Fig. 13. Sensitivity analysis of economic metrics with CO₂ penalty and discount rate impacts for (a) HRES1 (PV-Wind-Battery-Fuel cell-Grid) (b) HRES2 (PV-Battery-Fuel cell-Grid) (c) HRES3 (PV-Wind-Battery-Grid) (d) HRES4 (PV-Battery-Grid).

Table 8
Comparative analysis with the other studies.

Ref.	Location	PV	WT	H ₂	EZ	BAT	FC	HY	LCOE (\$/kWh)	LCOH (\$/kg)
[2]	Muscat, Oman	✓	✓			✓			0.0158	0.401
[17]	Northern Ghana					✓		✓	0.06	4.47
[26]	Northern-western region of Oman	✓	✓		✓				–	8.34–16.73
[29]	Egypt-Yemen	✓	✓		✓				0.010–0.012	2.35–2.70
[35]	Ludhiana, Indore, Mumbai, Chennai, Kolkata	✓		✓	✓		✓		0.41–0.48	3.00–3.22
[40]	Dhahran city, Saudi Arabia	✓	✓	✓	✓	✓			0.593	36.32
[41]	Doha Intl Airport, Qatar	✓	✓	✓					11.45	2.092
[42]	North-East, India	✓		✓					0.509–0.689	16.1–18.5
[43]	Izmir-Cesme, Turkey	✓	✓	✓					0.146–0.152	7.53–7.87
[44]	Jeju island, Korea			✓					0.03	4.2–9.54
[45]	Khuzestan province, Iran	✓	✓	✓					0.1155	4.59
[46]	Badakhshan, Afghanistan		✓	✓					0.0529–0.1135	3.88–10.82
[47]	South Africa		✓	✓					0.098–0.14	6.34–8.97
[48]	Al-Kharj, Saudi Arabia		✓	✓	✓				0.6208	9.34
This study	Cox's Bazar, Bangladesh	✓	✓	✓	✓	✓			0.0321	5.22

Note: WT = Wind Turbine, H₂ = Hydrogen, EZ = Electrolyzer, BAT = Battery, FC = Fuel cell, HY=Hydro.

potentially overlooking certain optimal configurations. Future studies should focus on integrating robust metaheuristic optimization techniques, to explore a wider range of potential solutions. Additionally, future research could assess off-grid HRES configurations for hydrogen production in remote regions, evaluating their feasibility for decentralized energy systems. Incorporating demand-side management strategies could further enhance the efficiency and cost-effectiveness of hydrogen production systems by better aligning energy supply with demand. These advancements would provide a more comprehensive

understanding of HRES capabilities for sustainable hydrogen production and renewable energy integration.

CRedit authorship contribution statement

Prangon Chowdhury: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Tasniah Islam:** Writing –

original draft, Visualization, Validation, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Ephraim Bonah Agyekum:** Writing – review & editing, Visualization, Validation, Supervision, Project administration, Methodology, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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